BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY

Computer Science and Engineering

Lab Report

If-else Problems

Microprocessor and Microcontroller Lab (CSE-326)

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Contents

1 Alphabetic order

1.0.1 Question

1. Take input two letters and print them in alphabetical order.

   Sample input/output screen:

   ```
   Enter the first letter: H
   Enter the second letter: A
   The letters in alphabetical order: AH
   ```

1.0.2 Solution

```
.MODEL SMALL
.STACK 100H

.DATA
STR  DB "Enter the first letter: $"
STR1 DB 10,13,"Enter the second letter: $"
STR2 DB 10,13,"The letters in alphabetical order: $"

.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    MOV AH, 9
    LEA DX, STR
    INT 21H

    MOV AH, 1
    INT 21H
    MOV BL, AL

    MOV AH, 9
    LEA DX, STR1
    INT 21H

    MOV AH, 1
    INT 21H
    MOV CL, AL

    MOV AH, 9
    LEA DX, STR2
    INT 21H

    CMP BL, CL
    JLE Order
```

```
    MOV AH, 2
    MOV DL, CL
    INT 21H

    MOV DL, BL
    INT 21H
    JMP Exit

Order:
    MOV AH, 2
    MOV DL, BL
    INT 21H

    MOV DL, CL
    INT 21H

Exit:
    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

Listing 1: Take input two letters and print them in alphabetical order.

Output

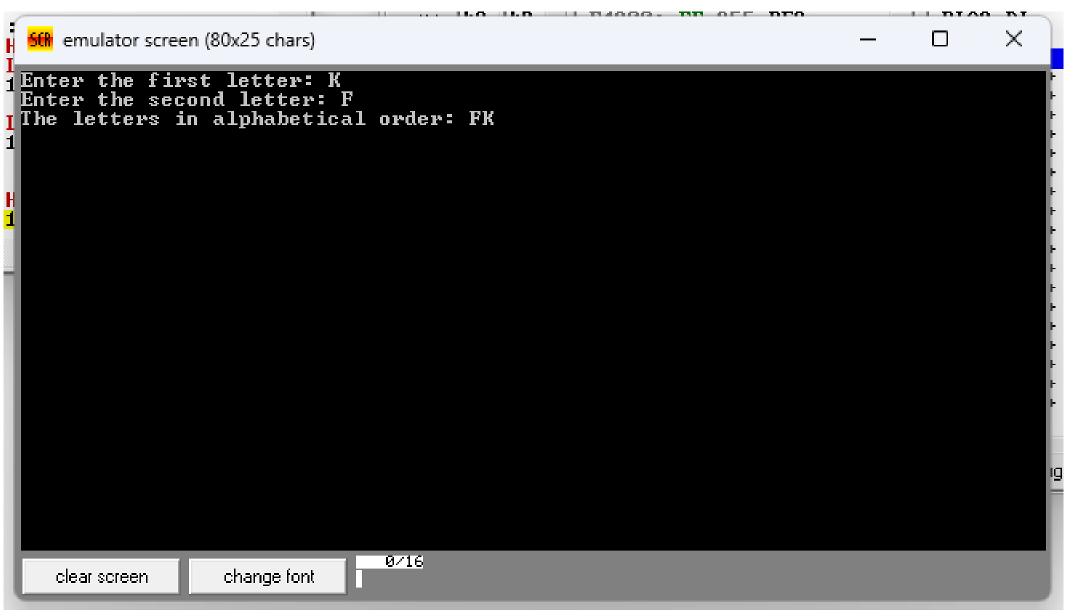

Figure 1: Task 1

“‘latex

2 Reverse alphabetic order

2.0.1 Question

1. Take input two letters and print them in reverse alphabetical order.

   Sample input/output screen:

   ```
   Enter the first letter: H
   Enter the second letter: A
   The letters in reverse alphabetical order: HA
   ```

2.0.2 Solution

```
.MODEL SMALL
.STACK 100H

.DATA
STR  DB "Enter the first letter: $"
STR1 DB 10,13,"Enter the second letter: $"
STR2 DB 10,13,"The letters in reverse alphabetical order: $"

.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    MOV AH, 9
    LEA DX, STR
    INT 21H

    MOV AH, 1
    INT 21H
    MOV BL, AL

    MOV AH, 9
    LEA DX, STR1
    INT 21H

    MOV AH, 1
    INT 21H
    MOV CL, AL

    MOV AH, 9
    LEA DX, STR2
    INT 21H

    CMP BL, CL
    JGE Order
```

```
    MOV AH, 2
    MOV DL, CL
    INT 21H

    MOV DL, BL
    INT 21H
    JMP Exit

Order:
    MOV AH, 2
    MOV DL, BL
    INT 21H

    MOV DL, CL
    INT 21H

Exit:
    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

Listing 2: Take input two letters and print them in reverse alphabetical order.

Output

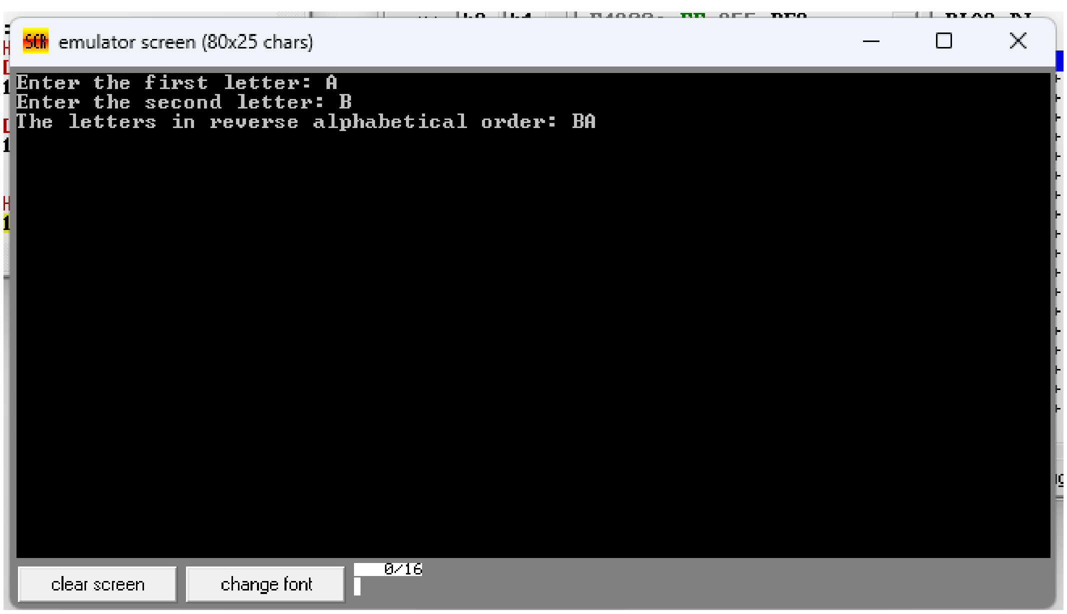

Figure 2: Task 2

3 Vowel or consonant

3.0.1 Question

1. Take letter input and check if it is a vowel or not.

   Sample input/output screen:

   ```
   Enter letter: A
   It is a vowel.

   (another example)
   Enter letter: K
   It is a consonant.
   ```

3.0.2 Solution

```
.MODEL SMALL
.STACK 100H

.DATA
STR  DB "Enter letter: $"
STR1 DB 10,13,"It is a vowel$"
STR2 DB 10,13,"It is a consonant$"

.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    MOV AH, 9
    LEA DX, STR
    INT 21H

    MOV AH, 1
    INT 21H
    MOV BL, AL

    CMP BL, 'A'
    JE Vowel
    CMP BL, 'E'
    JE Vowel
    CMP BL, 'I'
    JE Vowel
    CMP BL, 'O'
    JE Vowel
    CMP BL, 'U'
    JE Vowel

```

```
    CMP BL, 'a'
    JE Vowel
    CMP BL, 'e'
    JE Vowel
    CMP BL, 'i'
    JE Vowel
    CMP BL, 'o'
    JE Vowel
    CMP BL, 'u'
    JE Vowel

    MOV AH, 9
    LEA DX, STR2
    INT 21H
    JMP EXIT

Vowel:
    MOV AH, 9
    LEA DX, STR1
    INT 21H

EXIT:
    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

Listing 3: Take letter input and check if it is a vowel or not.

Output

(next page)

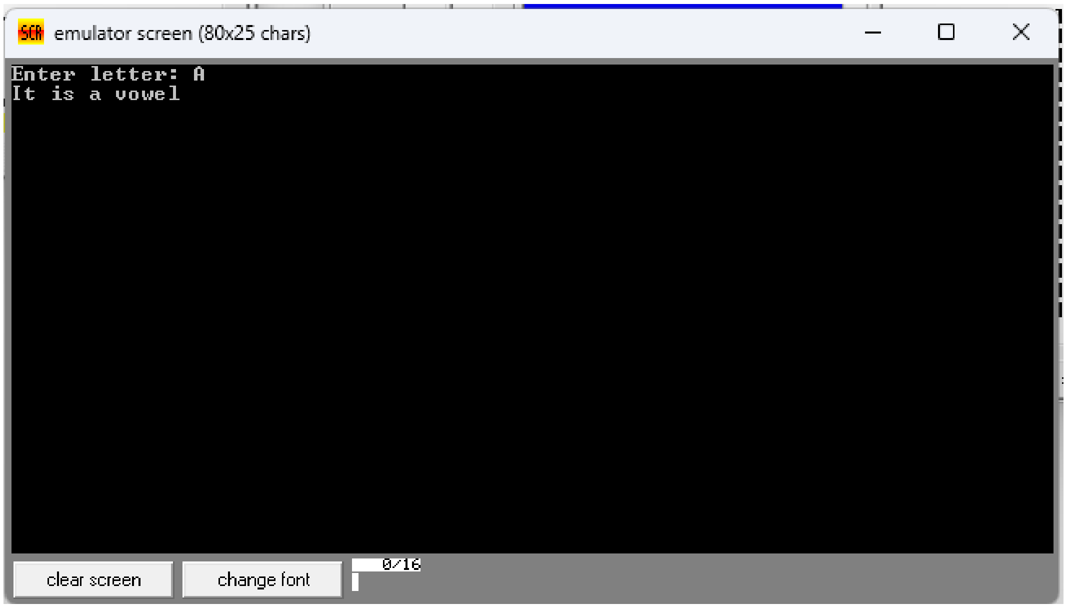

Figure 3: Task 3

4 Alphabet or digit

4.0.1 Question

1. Take a character input and check whether it is alphabet or digit.

   Sample input/output screen:

```
Enter character: A
Alphabet

(another example)
Enter character: 7
Digit
```

4.0.2 Solution

```
.MODEL SMALL
.STACK 100H

.DATA
STR  DB "Enter a character: $"
STR1 DB 10,13,"Digit$"
STR2 DB 10,13,"Alphabet$"
STR3 DB 10,13,"Invalid character$"

.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    MOV AH, 9
    LEA DX, STR
    INT 21H

    MOV AH, 1
    INT 21H
    MOV BL, AL

    ; check digit 0-9
    CMP BL, '0'
    JL CHECK_UPPER
    CMP BL, '9'
    JLE DIGIT

CHECK_UPPER:
    ; check uppercase A-Z
    CMP BL, 'A'
    JL INVALID
    CMP BL, 'Z'
    JLE ALPHABET

CHECK_LOWER:
    ; check lowercase a-z
    CMP BL, 'a'
    JL INVALID
    CMP BL, 'z'
    JLE ALPHABET

    JMP INVALID

DIGIT:
    MOV AH, 9
    LEA DX, STR1
    INT 21H
    JMP EXIT

```

```
ALPHABET:
    MOV AH, 9
    LEA DX, STR2
    INT 21H
    JMP EXIT

INVALID:
    MOV AH, 9
    LEA DX, STR3
    INT 21H

EXIT:
    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

Listing 4: Take a character input and check whether it is alphabet or digit.

Output

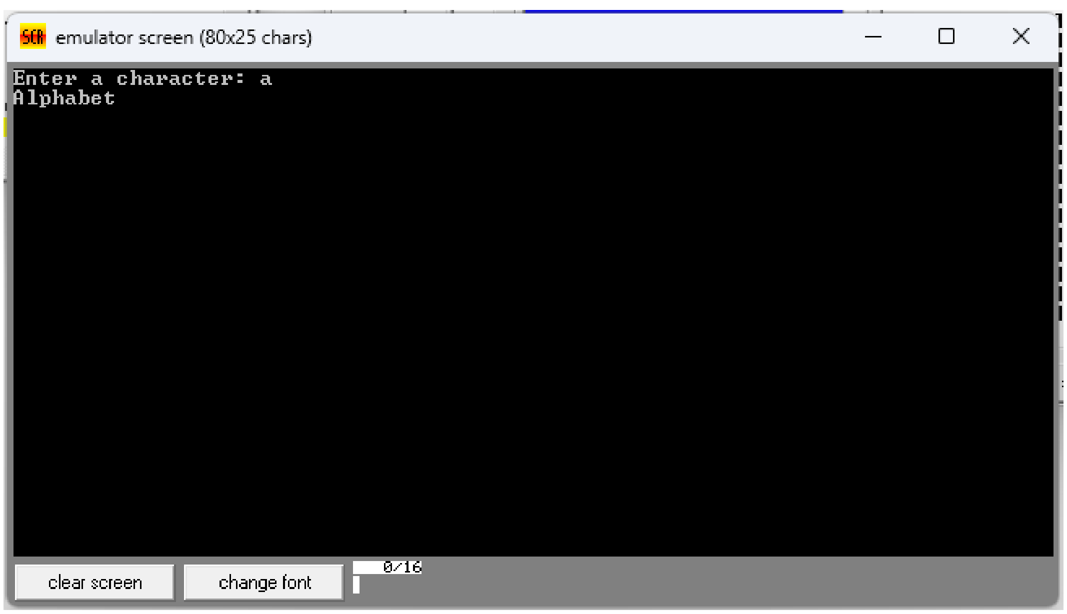

Figure 4: Task 4

5 Smallest among three digits

5.0.1 Question

1. Find out the smallest among 3 digits.

   Sample input/output screen:

   ```
   Enter first value: 8
   Enter second value: 1
   Enter third value: 6
   The smallest value is: 1
   ```

5.0.2 Solution

```
.MODEL SMALL
.STACK 100H

.DATA
STR  DB "Enter first digit: $"
STR1 DB 10,13,"Enter second digit: $"
STR2 DB 10,13,"Enter third digit: $"
STR3 DB 10,13,"Smallest: $"
STR4 DB 10,13,"Largest: $"

.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    MOV AH, 9
    LEA DX, STR
    INT 21H

    MOV AH, 1
    INT 21H
    MOV BL, AL

    MOV AH, 9
    LEA DX, STR1
    INT 21H

    MOV AH, 1
    INT 21H
    MOV CL, AL

    MOV AH, 9
    LEA DX, STR2
    INT 21H
```

```

    MOV AH, 1
    INT 21H
    MOV CH, AL

    ; assume BL = smallest, CL = middle, CH = largest

    CMP BL, CL
    JLE STEP1
    MOV DL, BL
    MOV BL, CL
    MOV CL, DL

STEP1:
    CMP CL, CH
    JLE STEP2
    MOV DL, CL
    MOV CL, CH
    MOV CH, DL

STEP2:
    CMP BL, CL
    JLE PRINT
    MOV DL, BL
    MOV BL, CL
    MOV CL, DL

PRINT:
    MOV AH, 9
    LEA DX, STR3
    INT 21H

    MOV AH, 2
    MOV DL, BL
    INT 21H

    MOV AH, 9
    LEA DX, STR4
    INT 21H

    MOV AH, 2
    MOV DL, CH
    INT 21H

EXIT:
    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

Listing 5: Find out the smallest among 3 digits.

Output

Figure 5: Task 5

[1] [2]

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